

Four options are given in each of the following questions fill the circle in front of that choice which you think is correct. Cutting or filling two or more circles is not allowed:

Q.1

- (i) The branch of mechanics which deals with the motion of objects is known as
☒ (a) dynamics (b) static (c) fluid mechanics (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (ii) The branch of applied mathematics that deals with the study of the motion of a body without taking into consideration how the motion is caused is known as
 (a) dynamics (b) static (c) fluid mechanics ☒ (d) kinematics
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (iii) _____ treats of the geometry of motion of bodies without taking into account their masses or the forces acting on them.
 (a) Kinetics (b) Static ☒ (c) Kinematics (d) Astrophysics
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (iv) The curve that is traced by the particle as the time progresses is called the
☒ (a) trajectory (b) velocity (c) acceleration (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (v) The instantaneous velocity of a particle is given by
 (a) $\lim_{\delta t \rightarrow 0} \frac{\delta^2 \vec{r}}{\delta t^2}$ (b) $\lim_{\delta t \rightarrow 0} \frac{\delta^3 \vec{r}}{\delta t^3}$ ☒ (c) $\lim_{\delta t \rightarrow 0} \frac{\delta \vec{r}}{\delta t}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)

- (vi) The instantaneous acceleration of a particle is given by
☒ (a) $\lim_{\delta t \rightarrow 0} \frac{\delta \vec{v}}{\delta t}$ (b) $\lim_{\delta t \rightarrow 0} \frac{\delta^2 \vec{v}}{\delta t^2}$ (c) $\lim_{\delta t \rightarrow 0} \frac{\delta^3 \vec{v}}{\delta t^3}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (vii) $\vec{v} =$
☒ (a) $\frac{d\vec{r}}{dt}$ (b) $\frac{d^2 \vec{r}}{dt^2}$ (c) $\frac{d^3 \vec{r}}{dt^3}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (viii) $\vec{a} =$
 (a) $\frac{d^3 \vec{v}}{dt^3}$ (b) $\frac{d^2 \vec{v}}{dt^2}$ ☒ (c) $\frac{d\vec{v}}{dt}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (ix) $\vec{a} =$
 (a) $\frac{d^3 \vec{r}}{dt^3}$ ☒ (b) $\frac{d^2 \vec{r}}{dt^2}$ (c) $\frac{d\vec{r}}{dt}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (x) If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{v} =$
☒ (a) $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$ (b) $\frac{d^2 x}{dt^2}\hat{i} + \frac{d^2 y}{dt^2}\hat{j}$ (c) $\frac{d^3 x}{dt^3}\hat{i} + \frac{d^3 y}{dt^3}\hat{j}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (xi) If $\vec{r} = x\hat{i} + y\hat{j}$, then $|\vec{v}| =$
 (a) $\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^2$ (b) $\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2$
☒ (c) $\sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2}$ (d) none of these ☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (xii) If $\vec{r} = x\hat{i} + y\hat{j}$, then $|\vec{a}| =$
 (a) $\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^2$ (b) $\sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2}$
☒ (c) $\sqrt{\left(\frac{d^2 x}{dt^2} \right)^2 + \left(\frac{d^2 y}{dt^2} \right)^2}$ (d) none of these ☐ (a) ☐ (b) ☐ (c) ☐ (d)

(xii) If $\vec{r} = x\hat{i} + y\hat{j}$, then $|\vec{a}| =$

(a) $\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^2$

(b) $\sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2}$

☒ (c) $\sqrt{\left(\frac{d^2x}{dt^2} \right)^2 + \left(\frac{d^2y}{dt^2} \right)^2}$

(d) none of these

☐ (a)

☐ (b)

☒ (c)

☐ (d)

(xiii) If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{a} =$

☒ (a) $\frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j}$

(b) $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

(c) $\frac{d^3x}{dt^3}\hat{i} + \frac{d^3y}{dt^3}\hat{j}$

(d) none of these

☒ (a)

☐ (b)

☐ (c)

☐ (d)

(xiv) The tangential component of velocity is the _____ of the velocity.

(a) square root of magnitude

(b) cube of magnitude

(c) square of magnitude

☒ (d) magnitude

☐ (a)

☐ (b)

☐ (c)

☒ (d)

(xv) The normal component of velocity is always

(a) negative

☒ (b) zero

(c) positive

(d) none of these

☐ (a)

☒ (b)

☐ (c)

☐ (d)

(xvi) At any instant the particle is moving in the direction of _____ to the path.

(a) normal

(b) opposite to normal

☒ (c) tangent

(d) none of these

☐ (a)

☐ (b)

☒ (c)

☐ (d)

(xvii) A particle is moving with a velocity $3\hat{i} + 7\hat{j}$, the tangential component of velocity is

(a) 58

☒ (b) $\sqrt{58}$

(c) 27

(d) 0

☐ (a)

☒ (b)

☐ (c)

☐ (d)

(xviii) A particle is moving with a velocity $3\hat{i} + 7\hat{j}$, normal component of velocity is

(a) 58

(b) $\sqrt{58}$

(c) 27

☒ (d) 0

☐ (a)

☐ (b)

☐ (c)

☒ (d)

(xix) At any time t the position of a particle moving in a plane can be specified by $(a\cos\omega t, a\sin\omega t)$, where a and ω are constants, the tangential component of velocity is

(a) 0

(b) $\sqrt{a\omega}$

☒ (c) $a\omega$

(d) $a^2\omega^2$

☐ (a)

☐ (b)

☒ (c)

☐ (d)

(xx) At any time t the position of a particle moving in a plane can be specified by $(a\cos\omega t, a\sin\omega t)$, where a and ω are constants, the normal component of velocity is

☒ (a) 0

(b) $\sqrt{a\omega}$

(c) $a\omega$

(d) $a^2\omega^2$

☐ (a)

☐ (b)

☐ (c)

☐ (d)

Q.2

- (i) The velocity has _____ along the normal to the path.
 (a) three components (b) two components
 (c) one component (d) no component (a) (b) (c) (d)
- (ii) If v is the magnitude of the velocity of a particle moving along a trajectory, then the tangential component of its acceleration is
 (a) $\frac{v^2}{\rho}$ (b) $\frac{dv}{dt}$ (c) $\frac{v^2}{\rho^2}$ (d) none of these
 (a) (b) (c) (d)
- (iii) The normal component of acceleration of a particle moving with the velocity of magnitude v is
 (a) $\frac{v^2}{\rho}$ (b) $\frac{dv}{dt}$ (c) $\frac{v^2}{\rho^2}$ (d) none of these
 (a) (b) (c) (d)
- (iv) If a particle is moving with uniform speed, then the tangential component of its acceleration is
 (a) negative (b) zero (c) positive (d) none of these
 (a) (b) (c) (d)
- (v) A particle is moving with a speed $v = t \sin t$, the tangential component of its acceleration is
 (a) $\sin t$ (b) $t \cos t$ (c) $t \sin t + \cos t$ (d) $t \cos t + \sin t$
 (a) (b) (c) (d)
- (vi) A particle is moving with a speed $v = t \sin t$, the tangential component of its acceleration at $t = 0$ is

- (a) 0 (b) 1 (c) 2 (d) 3
 (a) (b) (c) (d)
- (vii) A particle is moving with a speed $v = t \sin t$, the tangential component of its acceleration at $t = \frac{\pi}{2}$ is
 (a) 0 (b) 1 (c) 2 (d) 3
 (a) (b) (c) (d)
- (viii) A particle is moving with a speed $v = \int_0^t \cos x dx$, the tangential component of its acceleration is
 (a) $\sin t$ (b) $t \cos t$ (c) $t \sin t + \cos t$ (d) $\cos t$
 (a) (b) (c) (d)
- (ix) A particle is moving with a speed $v = e^t + t \cos t$, the tangential component of its acceleration at $t = 0$ is
 (a) 0 (b) 1 (c) 2 (d) 3
 (a) (b) (c) (d)
- (x) The direction of the radius vector is known as
 (a) radial direction (b) transverse direction
 (c) tangential direction (d) none of these (a) (b) (c) (d)
- (xi) The direction perpendicular to radius vector in the direction of increasing angle is called
 (a) radial direction (b) transverse direction
 (c) tangential direction (d) none of these (a) (b) (c) (d)
- (xii) The radial component of velocity is
 (a) $\dot{r}$ (b) $r\dot{\theta}$ (c) $\ddot{r}$ (d) $\dot{r}$
 (a) (b) (c) (d)
- (xiii) The transverse component of velocity is
 (a) $\dot{r}$ (b) $r\dot{\theta}$ (c) $\ddot{r}$ (d) $\dot{r}$
 (a) (b) (c) (d)

- (xiii) The transverse component of velocity is
 (a) $\dot{r}\dot{\theta}$ (b) $r\ddot{\theta}$ (c) $\ddot{r}$ (d) $\dot{r}$
 (a) (b) (c) (d)
- (xiv) The radial component of acceleration is
 (a) $\ddot{r} - r(\dot{\theta})^2$ (b) $\frac{1}{r} \frac{d}{dt}(r^2\dot{\theta})$ (c) $\frac{1}{r^2} \frac{d}{dt}(r^2\dot{\theta})$ (d) none of these
 (a) (b) (c) (d)
- (xv) The transverse component of acceleration is
 (a) $\ddot{r} - r(\dot{\theta})^2$ (b) $\frac{1}{r} \frac{d}{dt}(r^2\dot{\theta})$ (c) $\frac{1}{r^2} \frac{d}{dt}(r^2\dot{\theta})$ (d) none of these
 (a) (b) (c) (d)
- (xvi) The transverse component of acceleration is
 (a) $\ddot{r} - r(\dot{\theta})^2$ (b) $2\dot{r}\dot{\theta} + r\ddot{\theta}$ (c) $2\dot{r}\ddot{\theta}$ (d) none of these
 (a) (b) (c) (d)
- (xvii) The position of a particle moving along a path at any time t is given by $\vec{r} = \sin t \cos t \hat{i} + \sin^2 t \hat{j}$ and the polar angle is $\theta = 7t$. The radial component of velocity is
 (a) $\tan t$ (b) $\cot t$ (c) $\sin t$ (d) $\cos t$
 (a) (b) (c) (d)

- (xviii) The position of a particle moving along a path at any time t is given by $\vec{r} = \sin t \cos t \hat{i} + \sin^2 t \hat{j}$ and the polar angle is $\theta = 7t$. The transverse component of velocity is
 (a) $7 \sin t$ (b) $6 \cos t$ (c) $7 \sin t$ (d) $-\sin t$
 (a) (b) (c) (d)
- (xix) The position of a particle moving along a path at any time t is given by $\vec{r} = \cos^2 t \hat{i} + \cos t \sin t \hat{j}$ and the polar angle is $\theta = 6t$. The radial component of velocity is
 (a) $7 \sin t$ (b) $6 \cos t$ (c) $7 \sin t$ (d) $-\sin t$
 (a) (b) (c) (d)
- (xx) The position of a particle moving along a path at any time t is given by $\vec{r} = \cos^2 t \hat{i} + \cos t \sin t \hat{j}$ and the polar angle is $\theta = 6t$. The transverse component of velocity is
 (a) $7 \sin t$ (b) $6 \cos t$ (c) $7 \sin t$ (d) $-\sin t$
 (a) (b) (c) (d)

Short Questions

Q.3 Solve / write answers of the following short questions:

- (i) What is dynamics?
 (ii) What is kinematics?
 (iii) A particle is moving in such a way that its position at any time t is specified by

Four options are given in each of the following questions fill the circle in front of that choice which you think is correct. Cutting or filling two or more circles is not allowed:

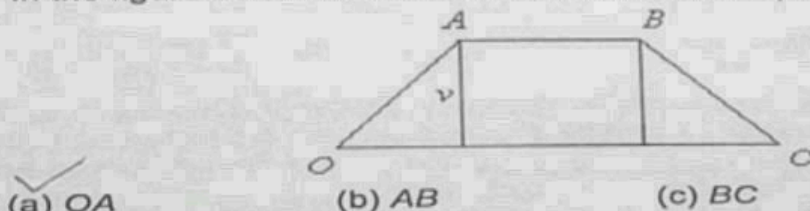
Q.1

- (i) The motion along a straight line is called
 (a) circular motion ☐ (b) ☒ rectilinear motion
 (c) projectile motion ☐ (d) none of these ☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (ii) If x is the distance of a particle from a fixed point at any time t , then the magnitude of the velocity is
☒ (a) $\frac{dx}{dt}$ (b) $\frac{d^2x}{dt^2}$ (c) $\left(\frac{dx}{dt}\right)^2$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (iii) If x is the distance of a particle from a fixed point at any time t , then the magnitude of the acceleration is
 (a) $\frac{dx}{dt}$ ☒ (b) $\frac{d^2x}{dt^2}$ (c) $\left(\frac{dx}{dt}\right)^2$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (iv) $v =$
 (a) $v^2 - u^2$ (b) $ut + \frac{1}{a}at^2$ ☒ (c) $u + at$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)

- (v) $x =$
 (a) $v^2 - u^2$ ☒ (b) $ut + \frac{1}{2}at^2$ (c) $u + at$ (d) none of these
- (vi) $2ax =$
☒ (a) $v^2 - u^2$ (b) $ut + \frac{1}{2}at^2$ (c) $u + at$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (vii) The distance travelled by a particle moving in a straight line with uniform acceleration a in the n th unit of time, where u is initial speed of the particle is
 (a) $u - \frac{1}{4}a(2n-1)$ (b) $u + \frac{1}{4}a(2n-1)$ ☒ (c) $u + \frac{1}{2}a(2n-1)$ (d) $u + a(2n-1)$
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (viii) The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in the 8th second is
 (a) $u - \frac{15}{2}a$ (b) $u + \frac{17}{2}a$ ☒ (c) $u + \frac{15}{2}a$ (d) $u - \frac{17}{2}a$
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (ix) The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in the 9th second is
 (a) $u - \frac{15}{2}a$ ☒ (b) $u + \frac{17}{2}a$ (c) $u + \frac{15}{2}a$ (d) $u - \frac{17}{2}a$
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (x) When the acceleration is a function of time only, we may write $a(t) =$
☒ (a) $\frac{d^2x}{dt^2}$ (b) $v \frac{dv}{dx}$ (c) $\frac{dx}{dt}$ (d) none of these
☐ (a) ☐ (b) ☐ (c) ☐ (d)
- (xi) When the acceleration is a function of time only, we may write $a(t) =$
 (a) $\frac{dx}{dt}$ ☒ (b) $v \frac{dv}{dx}$ ☒ (c) $\frac{dv}{dt}$ (d) none of these

- (xi) When the acceleration is a function of time only, we may write $a(t) =$
 (a) $\frac{dx}{dt}$ (b) $v \frac{dv}{dx}$ (c) $\frac{dv}{dt}$ (d) none of these
 (a) (b) (c) (d)
- (xii) When the acceleration is a function of velocity v only, we may write $a(v) =$
 (a) $\frac{d^2x}{dt^2}$ (b) $\frac{dv}{dt}$ (c) $\frac{dx}{dt}$ (d) none of these
 (a) (b) (c) (d)
- (xiii) When the acceleration is a function of velocity v only, we may write $a(v) =$
 (a) $\frac{d^2x}{dt^2}$ (b) $v \frac{dv}{dx}$ (c) $\frac{dx}{dt}$ (d) none of these
 (a) (b) (c) (d)
- (xiv) When the acceleration is a function of distance x only, we may write $a(x) =$
 (a) $\frac{dx}{dt}$ (b) $\frac{dv}{dt}$ (c) $v \frac{dv}{dx}$ (d) none of these
 (a) (b) (c) (d)
- (xv) When the acceleration is a function of distance x only, we may write $a(x) =$
 (a) $\frac{dx}{dt}$ (b) $\frac{dv}{dt}$ (c) $\frac{d^2x}{dt^2}$ (d) none of these
 (a) (b) (c) (d)

- (xix) For a constant velocity, the slope of the space-time curve
 (a) does not remain the same (b) varies
 (c) remains the same (d) none of these (a) (b) (c) (d)
- (xx) The space-time curve is a _____ for a constant velocity.
 (a) circle (b) straight line (c) parabola (d) hyperbola
 (a) (b) (c) (d)
- Q.2
- (i) The _____ of a particle moving in a straight line is given by the slope of the velocity-time curve.
 (a) acceleration (b) velocity (c) speed (d) none of these
 (a) (b) (c) (d)
- (ii) If the acceleration is constant, the velocity-time curve will be a _____ whose slope gives the acceleration.
 (a) parabola (b) circle (c) hyperbola (d) straight line
 (a) (b) (c) (d)
- (iii) The slope of the velocity-time curve of a particle moving in a straight line gives its
 (a) acceleration (b) velocity (c) speed (d) none of these
 (a) (b) (c) (d)
- (iv) The area under the velocity-time curve of a particle moving in a straight line gives the _____ by the particle.
 (a) acceleration (b) velocity (c) speed (d) distance
 (a) (b) (c) (d)
- (v) In the figure below the accelerated motion of the particle is along



(vi) In the figure below the uniform motion of the particle is along



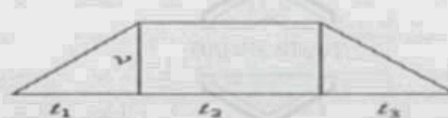
- (a) OA (b) AB (c) BC (d) none of these

(vii) In the figure below the retarded motion of the particle is along



- (a) OA (b) AB (c) BC (d) none of these

(viii) In the figure below the distance covered by the particle is



- (a) $\frac{1}{4}v(t_1 + t_2 + t_3)$ (b) $\frac{1}{2}v(t_1 + t_2 + t_3)$ (c) $\frac{1}{2}v(t_1 + 2t_2 + t_3)$ (d) $v(t_1 + t_2 + t_3)$

(ix) In the figure below the distance covered by the particle is



- (a) $\frac{1}{2}v(t_1 - t_2)$ (b) $\frac{1}{4}v(t_1 + t_2)$ (c) $\frac{1}{2}v(t_1 + t_2)$ (d) $v(t_1 + t_2)$

(x) A particle is said to execute Simple Harmonic Motion if it moves in a straight line so that its acceleration is always directed towards a fixed point in the line and is proportional to the _____ of the particle from the fixed point.
(a) square root of the distance (b) distance (c) square of the distance (d) none of these

(xi) The maximum distance covered by the particle executing SHM in either side of mean position is called

- (a) amplitude (b) time period (c) frequency (d) none of these

(xii) The time taken by the particle to complete one oscillation is called

- (a) amplitude (b) time period (c) frequency (d) none of these

(xiii) The time taken by the particle to complete one oscillation is called

- (a) amplitude (b) time period (c) frequency (d) none of these

(xiv) The particle executing SHM has _____ velocity at mean position.

- (a) minimum (b) zero (c) maximum (d) none of these

(xv) The time period of a particle executing SHM is

- (a) $\frac{2\pi}{\sqrt{\lambda}}$ (b) $\frac{\sqrt{\lambda}}{2\pi}$ (c) $\frac{\pi}{\sqrt{\lambda}}$ (d) $2\pi\sqrt{\lambda}$

(xvi) The frequency of a particle executing SHM is

- (a) $\frac{2\pi}{\sqrt{\lambda}}$ (b) $\frac{\sqrt{\lambda}}{2\pi}$ (c) $\frac{\pi}{\sqrt{\lambda}}$ (d) $2\pi\sqrt{\lambda}$

(xvii) The differential equation of a particle executing SHM is

- (a) $\frac{d^2x}{dt^2} + \lambda \frac{dx}{dt} = 0$ (b) $\frac{dx}{dt} + \lambda x = 0$ (c) $\frac{d^2x}{dt^2} + \lambda x = 0$ (d) none of these

(xviii) The projection of a particle moving along a circle executes

- (a) projectile motion (b) motion along a parabola (c) simple harmonic motion (d) none of these



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MTH404 - Dynamics (Quiz #1)

Question # 1 of 10 (Start time: 12:45:09 PM, 30 May 2023)

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

☐	maximum
☐	minimum
☐	none of these
☐	zero

MTH404 - Dynamics (Quiz #1)

Question # 2 of 10 (Start time: 12:45:49 PM, 30 May 2023)

If the displacement of a particle is given by

$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be -----.

Select the correct option

- | | |
|-----------------------|------|
| ☐ | $3t$ |
| ☐ | $6t$ |
| ☐ | 6 |
| ☐ | 0 |

MTH404 - Dynamics (Quiz #1)

Question # 3 of 10 (Start time: 12:47:06 PM, 30 May 2023)

Time period is reciprocal of-----

Select the correct option

☐	amplitude
☐	cycle
☒	frequency
☐	wavelength

MTH404 - Dynamics (Quiz #1)

Question # 4 of 10 (Start time: 12:47:38 PM, 30 May 2023)

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- | | |
|-----------------------|-----------------------------|
| ☐ | maximum |
| ☐ | neither maximum nor minimum |
| ☐ | none of these |
| ☐ | minimum |

Question # 5 of 10 (Start time: 12:48:12 PM, 30 May 2023)

Total Marks: 1

A particle is said to execute simple harmonic motion if it moves in a straight line so that its acceleration is always directed towards a fixed point in the line and is proportional to the ----- of the particle from the fixed point.

Select the correct option

☒	distance
☐	square root of the distance
☐	none of these
☐	square of the distance

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 12:48:47 PM, 30 May 2023)

The space-time curve is a ----- for a constant velocity.

Select the correct option

☐	circle
☐	parabola
☐	straight line
☐	hyperbola



Click to Save A

Question # 7 of 10 (Start time: 12:48:39 PM, 30 May 2023)

If the velocity of a particle is given by

$$v = 3t^2 + 4$$

then the acceleration of the particle at time 3 seconds will be _____

Select the correct option

☐	3
☐	18
☐	6
☐	0

Question # 8 of 10 (Start time: 12:50:59 PM, 30 May 2023)

if the acceleration is 5 m/s^2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option



hyperbola



straight line



parabola



circle

Click to En

MTH404 - Dynamics (Quiz #1)

Question # 9 of 10 (Start time: 12:51:51 PM, 30 May 2023)

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

☐	amplitude
☐	time period
☐	frequency
☐	none of these

Question # 10 of 10 (Start time: 12:52:12 PM, 30 May 2023)

Total Marks: 1

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. if its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm).

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|--------------|
| ☐ | $50\sqrt{2}$ |
| ☐ | $10\sqrt{3}$ |
| ☐ | $50\sqrt{3}$ |
| ☐ | 5 |

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 10 (start time: 01:40:51 PM, 30 May 2023)

Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

☒

frequency

☐

amplitude

☐

cycle

☐

wavelength

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 01:41:54 PM, 30 May 2023)

Total Mark

frequency is reciprocal of -----.

Select the correct option

☐

time period

☐

amplitude

☐

cycle

☐

wavelength

12 / 20



Click to Save Answer & Move to Next Question

Question # 4 of 10 (start time: 01:43:43 PM, 30 May 2023)

Total Marks: 1

If the velocity of the particle remains same throughout the motion, the slope of the space-time curve -----.

Select the correct option

☐ varies☐ None of these☒ Remains the same☐ Does not remain the same

14 / 20



Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 01:45:13 PM, 30 May 2023)

Total Marks: 1

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

☐

zero

☐

minimum

☐

maximum

☐

none of these



Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 01:47:24 PM, 30 May 2023)

Total Marks

If the acceleration is 5m/s^2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option

☐

straight line

☐

circle

☐

parabola

☐

hyperbola

[Click to Save Answer & Move to Next Question](#)

Question # 9 of 10 (start time: 01:48:23 PM, 30 May 2023)

Total Mark

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

- ☒ time period
- ☐ frequency
- ☐ none of these
- ☐ amplitude

Grand Quiz

Dynamics MCQ's (Lec 1 to 22)

1. The branch of mechanics which deals with the motion of object is known as
Ans: dynamics

2. The branch of applied mathematics that deals with study of motion of body without taking into consideration how the motion is caused is;
Ans: kinematics

3. Treat of the geometry of motion of bodies without taking into account their masses or the forces acting on them.
Ans: kinematics

4. The curve that is traced by a particle as the time progresses is called the
Ans: trajectory

5. The instantaneous velocity of particle is given by:
Ans: $\lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t}$

6. The instantaneous acceleration of a particle is given by.
Ans: $\lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$

7. $\vec{v} = ?$
Ans: $\frac{d\vec{r}}{dt}$

Made by Hajra

8- $\vec{a} = ?$

Ans $\frac{d\vec{v}}{dt}$ or $\frac{d^2\vec{r}}{dt^2}$

Made by Hajra

9- If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{v} = ?$

Ans $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

10- If $\vec{r} = x\hat{i} + y\hat{j}$, then $|\vec{v}| = ?$

Ans $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

11- If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{a} = ?$

Ans $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

12- The Tangential component of velocity is the magnitude of the velocity.

Ans magnitude

13- The normal component of velocity is always.

Ans zero

14- At any instant the particle is moving in the direction of -

Made by Hajra

to the path

Ans tangent

15- A particle is moving with a velocity $3\hat{i} + 7\hat{j}$ the tangential component of velocity is?

Ans 5.8

16- A particle is moving with a velocity $3\hat{i} + 7\hat{j}$, normal component of velocity is?

Ans zero "0"

17- At any time t the position of a particle moving in a plane can be specified by $(a \cos \omega t, a \sin \omega t)$ where a and ω are constant, the tangential component of velocity is

Ans $a\omega$

18- At any time t the position of a particle moving in a plane can be specified by $(a \cos \omega t, a \sin \omega t)$, where a & ω are constant the normal component of velocity is

~~zero~~ zero

Q#2

Made by Hajra

19- The velocity has _____ along the normal to the path

Ans- no component.

20- If v is the magnitude of velocity of a particle moving along a trajectory, then the tangential component of acceleration is

Ans $\frac{dv}{dt}$

21- The normal component of acceleration of a particle moving with the velocity magnitude v is

Ans $\frac{v^2}{r}$

22- If a particle is moving with uniform speed, then the tangential component of its acceleration is

Ans zero

23- A particle is moving with speed $v = v \sin t$, the tangential component of its acceleration is?

Ans: $+v \cos t + v \sin t$.

Made by Hajira

24. A particle is moving with a speed $v_0 + 5\sin t$, the tangential component of its acceleration at $t=0$ is

Aus Zero

25. A particle is moving with a speed $v_0 + 5\sin t$, the tangential component of its acceleration at $t = \frac{\pi}{2}$ is

Aus 1

26. A particle is moving with a speed $v_0 + \int_0^t \cos x dx$, the tangential component of its acceleration is

Aus $\cos t$

27. A particle is moving with a speed $v_0 e^t + t \cos t$, the tangential component of its acceleration at $t=0$ is

Aus 2

28. The direction of radius vector is known as

Aus Radial direction

Made by Hajra

29- The direction perpendicular to radius vector in the direction of increasing angle is called

Ans Transverse direction.

30- The radial component of velocity is

Ans $\dot{r}$

31- The Transverse component of velocity is

Ans $r\dot{\theta}$

32- The radial component of acceleration is

$$\ddot{r} - r(\dot{\theta})^2$$

33- The transverse component of acceleration is

Ans $\frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$ or also $2\dot{r}\dot{\theta} + r\ddot{\theta}$

34- The position of a particle moving along a path at any time t is given by $\vec{r} = \sin t \cos t \hat{i} + \sin^2 t \hat{j}$

Made by Hājra

40- If x is the distance of a particle from a fixed point at any time t , then the magnitude of velocity is

Ans $\frac{dx}{dt}$

41- If x is the distance of a particle from a fixed point at any time t , then the magnitude of acceleration is

Ans $\frac{d^2x}{dt^2}$

42- $v = ?$

Ans $u + at$

43- $x = ?$

Ans $ut + \frac{1}{2}at^2$

44- $2ax = ?$

Ans $v^2 - u^2$

45- The distance travelled by a

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46. particle moving in a straight line with uniform acceleration a in the n^{th} unit of time, where u is initial speed of the particle is

Ans $u + \frac{1}{2} a(2n-1)$

47. The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in the 8th second is

Ans $u + \frac{15}{2} a$

48. The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in 9th second is

Ans $u + \frac{17}{2} a$

49. When the acceleration is function of time only, we may write $a(t) =$

Ans $\frac{d^2x}{dt^2}$

50. When the acceleration is function of velocity only, we may write $a(v) =$

Ans $v \frac{dv}{dt}$ also $v \frac{dv}{dx}$

Made by Hajra

51- When the acceleration is a function of distance x only, we may write $a(x)$.

Ans $v \frac{dv}{dx}$ or also $\frac{d^2x}{dt^2}$

52- The slope of tangent to the curve $y = e^x \sin x$ is $\frac{dy}{dx} =$

Ans $e^x (\cos x + \sin x)$

53- The slope of tangent to the curve $y = e^x \sin x$ at $x=0$ is

Ans 1

54- The velocity of a particle at a distance x is equal to the slope of the space-time curve at a point whose coordinate is x .

Ans tangent.

55- For a constant velocity, the slope of space-time curve is

Ans remains the same

Made by Hajra

56. The space-time curve is a
for a constant velocity

Ans straight line.

57. The slope of a particle moving
in a straight line is given
by the slope of the velocity-time
curve.

Ans acceleration.

58. If the acceleration is constant,
the velocity-time curve will be
a straight line whose slope gives the
acceleration.

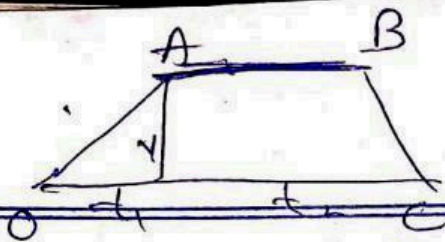
Ans straight line.

59. The slope of a velocity-time
curve of a particle moving in a
straight line gives its.

Ans acceleration.

60. The area under the velocity-time
curve of a particle moving in
a straight line gives the distance
by the particle.

Ans distance.



Made by Hajra

61. OA = accelerated motion
BC = ~~retard~~ retarded motion

$$\frac{1}{2} v (t_1 + t_2) = \text{distance covered by particle}$$

Def of S.H.M (imp).

62. The maximum distance covered by the particle executing S.H.M in either side of mean position is called

Ans. amplitude

63. The time taken by the particle to complete one oscillation is called

Ans. Time period

64. The particle executing S.H.M has velocity at mean position

Ans. maximum

65. The time period of a particle executing S.H.M is

Ans. $\frac{2\pi}{\omega}$

Made by Hajra

66. The frequency of particle executing S.H.M is

Ans $\frac{1}{T}$
 2π

67. The projection of a particle moving along a circle executes.

Ans:- Simple harmonic motion.

Q.1:-

68. If a force acts at a time t on a particle of mass m moving with velocity v , then the equation of motion according to Newton's second law is

Ans $F = \frac{d}{dt}(mv)$

69. A body of mass $m = 5g$ is executing S.H.M about a point O with amplitude $10cm$. If its max velocity is $100cm/sec$. Its velocity will be $50cm/sec$ at a distance

Ans $5\sqrt{3}$

Made by Hajra

70. A particle of mass 4 kg is in motion under the action of variable force F and is moving under velocity $3t^2$ then the force acting at time "2" second is.

Ans 48 N .

71. In a conservative field of force the total energy of particle ——— throughout the motion.

Ans remains constant.

72. A particle of mass 3 kg is moving with velocity $4i + 5j$. The force F acting on the particle will be.

Ans $12i$

73. At mean position of pendulum the potential energy of pendulum is.

Ans minimum

74. If a particle of mass 4 kg is moving with speed of 3 ms^{-1} then the kinetic energy of particle is.

Ans 18 J

$$\text{Work} = F \times d$$

$$= 20 \times 3$$

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75. Force acting on a body change its kinetic energy, direction of motion, speed, all

Ans all

76. If a particle is moving along a circle of radius a with center O . Which of the following is true about transverse component of force in polar.

$$F_{\theta} = m a \ddot{\theta}$$

77. During S.H.M the kinetic energy of body at Mean position is

Ans Maximum (because v is max).

78. Transverse component of force in polar coordinate.

$$F_{\theta} = m (2 \dot{r} \ddot{\theta} + r \ddot{\theta})$$

79. If x is displacement of particle from a fixed point at any time t and is given by $x = 5t^2$. Then expression for acceleration of

Made by Hajra

of particle is

Ans

10

80- which of the following is true for a force F to be conservative

Ans

$\text{Curl } F = 0$

81. The rate of change of a particle about a point O is equal to the torque about O of the force on the particle

Ans

angular momentum

82. If the mass of the particle is doubled, then the gravitational force is earth's gravitational field is

Ans

doubled

83. If the velocity of the particle is $\sqrt{3t^2 + 4}$ then the acceleration of particle at times "3" seconds will be

Ans

18

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84. Direction of momentum is direction of object's?

Ans. Velocity.

85. Radial component of force in polar coordinates.

Ans. $F_r = m(\ddot{r} - r(\dot{\theta})^2)$

86. Rate of work done to the particle is

Ans. Power



BC210202095: MUQADAS BATOOL

Time Left 72
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 10 of 10 (start time: 11:02:53 AM, 06 June 2023)

Total Marks: 1

If a particle is moving along a circle of radius 'a' with Center at O, then which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

Reveal Math Equations

☐	$F_\theta = m\ddot{\theta}$
☒	$F_\theta = ma\ddot{\theta}$
☐	None of these
☐	$F_\theta = m\dot{\theta}$

Click to Save Answer & Move to Next Question



BC210202095: MUQADAS BATOOL

Time Left 43
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 9 of 10 (Start time: 11:01:50 AM, 06 June 2023)

Total Marks: 1

If force of magnitude 10N is acting on the particle of mass 5kg, then the acceleration produced in the particle is-----.

Select the correct option

☐	15m/s ²
☐	10m/s ²
☒	2m/s ²
☐	5m/s ²



Click to Save Answer & Move to Next Question

11:01 AM

4G 65 B/s 83

Quiz
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BC210202095: MUQADAS BATOOL

Time Left 71
sec (s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 8 of 10 (Start time: 11:01:22 AM, 06 June 2023)

Total Marks: 1

Which of the following is true about the radial component of a force in polar coordinates?

Select the correct option

Reload Math Equations

☒	$F_r = m(\ddot{r} - r(\dot{\theta})^2)$
☐	None of these
☐	$F_r = m(\ddot{r} + r(\dot{\theta})^2)$
☐	$F_r = m(r(\dot{\theta})^2 - \ddot{r})$

Click to Save Answer & Move to Next Question

11:01 AM

4G 489 B/s 83

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left

81
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 7 of 10 (Start time: 11:01:08 AM, 06 June 2023)

Total Marks: 1

If the mass of the particle is doubled, then the gravitational force in the earth's gravitational field -----.

Select the correct option

☐	none of these
☐	remains same
☒	is doubled
☐	is trebled

Click to Save Answer & Move to Next Question

11:01 AM

4G 55 B/s 83

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 75
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 6 of 10 (Start time: 11:00:47 AM, 06 June 2023)

Total Marks: 1

The rate of change of ----- of a particle about a point O is equal to the torque about O of the force on the particle.

Select the correct option

☐	angular velocity
☐	none of these
☒	angular displacement
☐	angular momentum

Click to Save Answer & Move to Next Question

11:00 AM

4G 102 B/s 83



Quiz
vulms.vu.edu.pk



BC210202095: MUQADAS BATOOL

Time Left 69
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 5 of 10 (Start time: 11:00:14 AM, 06 June 2023)

Total Marks: 1

Which of the following is true for a force F to be conservative?

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☐ | $\text{div } F = 0$ |
| ☐ | $\text{grad } F = 0$ |
| ☒ | $\text{Curl } F = 0$ |
| ☐ | None of these |

Click to Save Answer & Move to Next Question

11:00 AM

4G 226 B/s 83

Quiz
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BC210202095: MUQADAS BATOOL

Time Left 72
sec (s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 4 of 10 (Start time: 10:59:50 AM, 06 June 2023)

Total Marks: 1

A particle of mass 3kg is moving with velocity $4t\mathbf{i} + 5t\mathbf{j}$. The force $\mathbf{F}$ acting on the particle will be-----

Select the correct option

☐	$4\mathbf{i} + 5\mathbf{j}$
☒	$12\mathbf{i}$
☐	0
☐	$12\mathbf{j}$

Click to Save Answer & Move to Next Question

10:59 AM

4G 24 B/s 83

Quiz
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BC210202095: MUQADAS BATOOL

Time Left 61
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 3 of 10 (Start time: 10:59:11 AM, 06 June 2023)

Total Marks: 1

If a particle of mass 4kg is in motion under the action of variable force F and is moving under velocity $3t^2$, then the force acting at time 2 seconds is -----.

Select the correct option

☐	12N
☐	24N
☒	48N
☐	7N

Click to Save Answer & Move to Next Question

10:59 AM

4G 24 B/s 83

Quiz
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BC210202095: MUQADAS BATOOL

Time Left 69
s.00 (s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 2 of 10 (Start time: 10:58:34 AM, 06 June 2023)

Total Marks: 1

If a force F acts at time t on a particle of mass m moving with velocity v , then the equation of motion, according to Newton's second law, is given by-----.

Select the correct option

Reload Math Equations

☐	$\vec{F} = m\vec{v}^2$
☐	none of these
☐	$\vec{F} = m\vec{v}$
☒	$\vec{F} = \frac{d}{dt}(m\vec{v})$

Click to Save Answer & Move to Next Question

10:58 AM

4G 94 B/s 83

Quiz
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BC210202095: MUQADAS BATOOL

Time Left 53
sec(s)

MTH404 - Dynamics (Quiz#2)

Quiz Start Time: 10:57 AM, 06 June 2023

Question # 1 of 10 (Start time: 10:57:30 AM, 06 June 2023)

Total Marks: 1

If force of magnitude 20N is applied to the particle and it covers a distance of 3m in the same direction as that of applied force, then the magnitude of work done on the particle is-----.

Select the correct option

☒	60J
☐	3J
☐	20J
☐	23J

Click to Save Answer & Move to Next Question